

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 8: Circling Approach Procedures

Syllabus Source: CAE Oxford (pp. 207-214) | Pass Target: >= 90% AviationExam

1. DEFINITION & OPERATIONAL CRITERIA

A circling approach is the visual phase of an instrument approach to bring an aircraft into position for landing on a runway which is not suitably located for a straight-in approach. A straight-in approach is considered unacceptable if the final approach track alignment differs from the runway centerline by MORE THAN 30°, or if the final descent gradient exceeds standard criteria (6.5% for Cat A/B, 6.1% for Cat C/D).

[*] OFFICIAL DEFINITION - Visual Reference Requirement in Circling

Visual reference to the runway environment (threshold, runway markings, or lights) must be MAINTAINED CONTINUOUSLY throughout the circling maneuver. The aircraft must remain at or above the circling MDA/H until positioned to make a normal landing on the intended runway.

2. CIRCLING AREA RADII & OBSTACLE CLEARANCE (MOC)

The circling area is constructed by drawing arcs from the threshold of each usable runway. The radius of the arc depends on the aircraft category speed (bank angle 20° or rate 3°/s):

Category	Max Circling Speed	Circling Radius from Thresholds	Minimum Obstacle Clearance (MOC)
Cat A	100 kt	1.68 NM (3.11 km)	90 m (295 ft)
Cat B	135 kt	2.66 NM (4.93 km)	90 m (295 ft)
Cat C	180 kt	4.20 NM (7.78 km)	120 m (394 ft)
Cat D	205 kt	5.28 NM (9.78 km)	120 m (394 ft)
Cat E	240 kt	6.94 NM (12.85 km)	150 m (492 ft)

3. DESCENT BELOW CIRCLING MDA/H & MISSED APPROACH

Descent below circling MDA/H is permitted ONLY when: 1. Required visual reference to the runway environment is established and maintained; 2. The runway threshold is in sight to the pilot; 3. The aircraft is in a position from which a normal descent to landing can be made on the runway using normal maneuvers.

[!] EXAM TRAP (AVIATIONEXAM) - Loss of Visual Reference While Circling (Guaranteed Exam Question)

If visual reference is lost at any time while circling to land: 1. The pilot must initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY. 2. Establish the aircraft overhead the aerodrome to ensure terrain clearance. 3. Join and follow the published missed approach procedure for the INSTRUMENT RUNWAY initially used.

4. PROHIBITED CIRCLING SECTORS & TERRAIN BUFFERS

Where prominent obstacles or terrain exist in a specific sector around an aerodrome, visual maneuvering may be prohibited in that sector (e.g. 'Circling prohibited North of RWY 09/27'). - The pilot must strictly contain all circling flight within the permitted visual sector. - If cleared to circle, turning into a prohibited sector constitutes an immediate breach of ATC clearance and terrain clearance.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: When is an instrument approach classified as a circling approach?

- [A]: Whenever the wind on landing runway exceeds 15 kt.
- [B]: When final approach track alignment differs by more than 30° from runway centerline.
- [C]: Whenever the pilot requests a visual traffic circuit.
- [D]: Only when the cloud base is below 1,000 ft.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under PANS-OPS, if final approach track differs by > 30° from runway centerline, only circling minima are authorized.

Q2: What is the circling area radius for a Category C aircraft?

- [A]: 1.68 NM
- [B]: 2.66 NM
- [C]: 4.20 NM
- [D]: 5.28 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Cat C circling radius is 4.20 NM from runway thresholds (based on 180 kt max speed).

Q3: What is the Minimum Obstacle Clearance (MOC) provided in a Category C circling area?

- [A]: 90 m (295 ft)
- [B]: 120 m (394 ft)
- [C]: 150 m (492 ft)
- [D]: 300 m (984 ft)

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cat A and B have 90 m MOC; Cat C and D have 120 m MOC; Cat E has 150 m MOC.

Q4: If visual reference is lost while circling to land on runway 27 after an instrument approach on runway 09, w

- [A]: Turn immediately towards runway 09 and descend.
- [B]: Make an immediate climbing turn towards runway 27 and join the missed approach for runway 09.
- [C]: Continue visually on downwind and attempt landing.
- [D]: Climb straight ahead on present heading to MSA.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The pilot must make an immediate climbing turn towards the landing runway (rwy 27) and establish overhead before following the missed approach for runway 09.

Q5: May a pilot descend below the circling MDA/H while still on the downwind leg?

- [A]: Yes, if the runway lights are visible
- [B]: NO, the aircraft must remain at or above circling MDA/H until aligned and in a normal descent position
- [C]: Yes, if cleared by ATC
- [D]: Only in daylight VMC

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Descent below circling MDA/H is prohibited until the aircraft is in a position from which a normal descent to landing can be executed.